Jingfeng Wu

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Academic Experience

University of California, Berkeley — Berkeley, US
Postdoctoral Fellow — 2023 - Onwards

- Advisors: Peter Bartlett and Bin Yu
- Part of the NSF/Simons Collaboration on the Theoretical Foundations of Deep Learning

Education

Johns Hopkins University — Baltimore, US
Ph.D. in Computer Science — 2019 - 2023

- Advisor: Vladimir Braverman
- Thesis: "Risk Convergence and Algorithmic Regularization of Discrete-Stepsize (Stochastic) Gradient Descent"

Peking University — Beijing, CN
M.S. in Applied Mathematics — 2016 - 2019
B.S. in Mathematics and Applied Mathematics — 2012 - 2016

Industrial Experience

Google Research — Seattle, US
Research Intern — 2022 Summer

- Mentors: Wennan Zhu and Peter Kairuoz

Baidu Research — Beijing, CN
Research Intern — 2018 Winter

- Mentor: Haoyi Xiong

Research Interests

Deep Learning Theory, Algorithms, Machine Learning, Optimization, Statistical Learning Theory

Conference Papers (asterisk * indicates equal contributions)

[1] **J. Wu**, D. Zou, Z. Chen, V. Braverman, Q. Gu, and P. L. Bartlett. "How Many Pretraining Tasks Are Needed for In-Context Learning of Linear Regression?" In: *International Conference on Learning Representations (ICLR)* (2024).

[2] X. Li, Y. Deng, **J. Wu**, D. Zhou, and Q. Gu. "Risk Bounds of Accelerated SGD for Overparameterized Linear Regression". In: *International Conference on Learning Representations (ICLR)* (2024).

[3] **J. Wu**, V. Braverman, and J. D. Lee. "Implicit Bias of Gradient Descent for Logistic Regression at the Edge of Stability". In: *Advances in Neural Information Processing Systems (NeurIPS)* (2023).

[4] **J. Wu**, W. Zhu, P. Kairouz, and V. Braverman. "Private Federated Frequency Estimation: Adapting to the Hardness of the Instance". In: *Advances in Neural Information Processing Systems (NeurIPS)* (2023).

[5] H. Li*, **J. Wu***, and V. Braverman. "Fixed Design Analysis of Regularization-Based Continual Learning". In: *Conference on Lifelong Learning Agents (CoLLAs)* (2023).

[6] **J. Wu***, D. Zou*, Z. Chen*, V. Braverman, Q. Gu, and S. M. Kakade. "Finite-Sample Analysis of Learning High-Dimensional Single ReLU Neuron". In: *International Conference on Machine Learning (ICML)* (2023).

[7] **J. Wu***, D. Zou*, V. Braverman, Q. Gu, and S. M. Kakade. "The Power and Limitation of Pretraining-Finetuning for Linear Regression under Covariate Shift". In: *Advances in Neural Information Processing Systems (NeurIPS)* (2022).

[8] D. Zou*, **J. Wu***, V. Braverman, Q. Gu, and S. M. Kakade. "Risk Bounds of Multi-Pass SGD for Least Squares in the Interpolation Regime". In: *Advances in Neural Information Processing Systems (NeurIPS)* (2022).

[9] **J. Wu***, D. Zou*, V. Braverman, Q. Gu, and S. M. Kakade. "Last Iterate Risk Bounds of SGD with Decaying Stepsize for Overparameterized Linear Regression". In: *International Conference on Machine Learning (ICML)* (2022).

[10] **J. Wu**, V. Braverman, and L. F. Yang. "Gap-dependent Unsupervised Exploration for Reinforcement Learning". In: *International Conference on Artificial Intelligence and Statistics (AISTATS)* (2022).

[11] D. Zou*, **J. Wu***, V. Braverman, Q. Gu, D. P. Foster, and S. M. Kakade. "The Benefits of Implicit Regularization from SGD in Least Squares Problems". In: *Advances in Neural Information Processing Systems (NeurIPS)* (2021).

[12] **J. Wu**, V. Braverman, and L. F. Yang. "Accommodating Picky Customers: Regret Bound and Exploration Complexity for Multi-Objective Reinforcement Learning". In: *Advances in Neural Information Processing Systems (NeurIPS)* (2021).

[13] H. Li, A. Krishnan, **J. Wu**, S. Kolouri, P. K. Pilly, and V. Braverman. "Lifelong Learning with Sketched Structural Regularization". In: *Asian Conference on Machine Learning (ACML)* (2021).

[14] Z. Yu, C. Hu, **J. Wu**, X. Sun, V. Braverman, M. Chowdhury, Z. Liu, and X. Jin. "Programmable Packet Scheduling with a Single Queue". In: *ACM Special Interest Group on Data Communication (SIGCOMM)* (2021).

[15] D. Zou*, **J. Wu***, V. Braverman, Q. Gu, and S. M. Kakade. "Benign Overfitting of Constant-Stepsize SGD for Linear Regression". In: *Conference on Learning Theory (COLT)* (2021).

[16] **J. Wu**, D. Zou, V. Braverman, and Q. Gu. "Direction Matters: On the Implicit Bias of Stochastic Gradient Descent with Moderate Learning Rate". In: *International Conference on Learning Representations (ICLR)* (2021).

[17] J. You, **J. Wu**, X. Jin, and M. Chowdhury. "Ship Compute or Ship Data? Why not Both?" In: *USENIX Symposium on Networked Systems Design and Implementation (NSDI)* (2021).

[18] Z. Yu, **J. Wu**, V. Braverman, I. Stoica, and X. Jin. "Twenty Years After: Hierarchical Core-Stateless Fair Queueing." In: *USENIX Symposium on Networked Systems Design and Implementation (NSDI)* (2021).

[19] **J. Wu**, V. Braverman, and L. F. Yang. "Obtaining Adjustable Regularization for Free via Iterate Averaging". In: *International Conference on Machine Learning (ICML)* (2020).

[20] **J. Wu**, W. Hu, H. Xiong, J. Huan, V. Braverman, and Z. Zhu. "On the Noisy Gradient Descent that Generalizes as SGD". In: *International Conference on Machine Learning (ICML)* (2020).

[21] B. Yu*, **J. Wu***, J. Ma, and Z. Zhu. "Tangent-Normal Adversarial Regularization for Semi-Supervised Learning". In: *Conference on Computer Vision and Pattern Recognition (CVPR)* (2019).

[22] Z. Zhu*, **J. Wu***, B. Yu, L. Wu, and J. Ma. "The Anisotropic Noise in Stochastic Gradient Descent: Its Behavior of Escaping from Sharp Minima and Regularization Effects". In: *International Conference on Machine Learning (ICML)* (2019).

Journal Papers (asterisk * indicates equal contributions)

[23] D. Zou*, **J. Wu***, V. Braverman, Q. Gu, and S. M. Kakade. "Benign Overfitting of Constant-Stepsize SGD for Linear Regression". In: *Journal of Machine Learning Research (JMLR)* (2023).

Professional Services

- **Conference Reviewer**
  - International Conference on Machine Learning (ICML), 2020 - 2024
  - Conference on Neural Information Processing Systems (NeurIPS), 2020 - 2023
  - International Conference on Learning Representations (ICLR), 2021 - 2024
  - International Conference on Artificial Intelligence and Statistics (AISTATS), 2021 - 2023
  - Conference on Uncertainty in Artificial Intelligence (UAI), 2023
- **Conference Program Committee Member**
  - AAAI Conference on Artificial Intelligence (AAAI), 2021 - 2023
- **Journal Reviewer**
  - IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
  - Journal of Machine Learning Research (JMLR)
  - Transactions on Machine Learning Research (TMLR)
  - Applied Probability Journals

Invited Talks

- **Statistics Seminar** at UC Davis — Octorber 2023
  "New Insights about SGD: Stepsize, Risk Convergence, and Implicit Regularization"
  - Host: Xiao Hui Tai
- **MoDL Collaboration Meeting** at Toyota Technological Institute at Chicago — May 2023
  "Implicit Bias of Gradient Descent for Logistic Regression at the Edge of Stability"
  - Hosts: Sam Buchanan, Omar Montasser, and Gal Vardi
- **Algorithms and ML Seminar** at Rice University — January 2023
  "The Implicit Regularization of SGD in Least Square Problems and Beyond"
  - Host: Anastasios Kyrillidis
- **Deep Learning Theory Seminar** at Princeton University — December 2022
  "A Fine-Grained Characterization for the Implicit Regularization of SGD in Least Square Problems"
  - Host: Jason D. Lee
- **Dynamical System Seminar** at Georgia Institute of Technology — November 2022
  "A Fine-Grained Characterization for the Implicit Regularization of SGD in Least Square Problems"
  - Host: Molei Tao
- **Learning Theory Seminar** at Google Research — August 2022
  "A Fine-Grained Characterization for the Implicit Regularization of SGD in Least Square Problems"
  - Host: Mehryar Mohri
- **Math Machine Learning Seminar** at MPI MiS and UCLA — June 2022
  "A Fine-Grained Characterization for the Implicit Regularization of SGD in Least Square Problems"
  - Host: Guido Montufar

Teaching

- **Teaching Assistant** at Johns Hopkins University — Spring 2023
  Machine Learning: Deep Learning

Awards

- University of Chicago and UCSD Rising Stars in Data Science — 2023
- Johns Hopkins University MINDS Summer Data Science Fellowship — 2021
- ICML Best Reviewers (Top 10%) — 2021